1

로그 미분법 ($\frac{f'(x)}{f(x)} = (\ln|f(x)|)'$)

→ 함수의 곱, 몫이 복잡하게 얽혀있을 때

지수와 밑 모두 변수 있을 때 ($f(x)^{g(x)}$ / x^x ...)

$f'(x) = f(x)\cdot\{\ln|f(x)|\}'$

$\{2\ln(x^3+2)+4x+\ln(\cos(5\tan x)) - \frac{1}{2}\ln(x^3+1)\}'$

$= \frac{6x^2}{x^3+2} + 4 + \frac{-\sin(5\tan x)}{\cos(5\tan x)}\cdot 5\sec^2 x - \frac{1}{2}\cdot\frac{3x^2}{x^3+1}$

$f'(0) = f(0)\cdot(0+4+0-0) = 16$

$4\cdot1\cdot1$

2

$\sin^{-1}\frac{\sqrt{3}}{2} + \cos^{-1}(\cos\frac{5}{9}\pi) = -\frac{\pi}{9}$

$\frac{\pi}{3}$, $-\frac{4}{9}\pi$, $\pi - \frac{4}{9}\pi$

3

$(\alpha,\beta) \Rightarrow y-\beta = -a(x-\alpha)$

$F: x^2y + \tan(\ln(1+xy)) - \cos(xy^2) - 3y = 0$.

$\frac{dy}{dx} = -\frac{2xy + \frac{1}{1+\ln^2(1+xy)}\cdot\frac{y}{1+xy} + y^2\sin(xy^2)}{x^2 + \frac{1}{1+\ln^2(1+xy)}\cdot\frac{x}{1+xy} + 2xy\cdot\sin(xy^2) - 3}\Big|_{(0,-\frac{1}{3})} = -\frac{1}{9}$

$\therefore y = -\frac{1}{9}x - \frac{1}{3}$

4

$\frac{dt}{dx} = \frac{1}{x}$ $dx = x\,dt$ $x = \frac{dx}{dt}$ $dx = x\,dt$ $\ln e = 1$ $\ln 1 = 0$

$\ln x \triangleq t \Rightarrow dx = e^t dt \Rightarrow \int_0^1 t^2 e^{2t} dt$.

$\therefore \int_1^e x(\ln x)^2 dx$

$= \int_0^1 e^t \times t^2 \times \frac{x}{e^t} x\,dt = \int_0^1 e^{2t} t^2 dt$

$2t \triangleq u \Rightarrow dt = \frac{1}{2}du$

$\frac{1}{2}\cdot\frac{1}{4}\int_0^2 u^2 e^u du = \frac{1}{8}[u^2e^u - 2ue^u + 2e^u]_0^2 = \frac{1}{8}\{(4e^2 - 4e^2 + 2e^2) - (2)\}$ ⇒ 로다삼지

$= \frac{e^2-1}{4}$

도표적분법

$[+u^2e^u - 2ue^u + 2e^u - 0\times e^u]_0^2$

$= \frac{1}{8}\{(4e^2 - 4e^2 + 2e^2) - (2)\}$

$= \frac{e^2-1}{4}$

5

$\bar{f} = \frac{1}{1-0}\int_0^1 f(x)dx$.

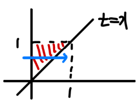
$\int_0^1\int_x^1 \cos(t^2)\,dt\,dx$

x의 범위: $0 \le x \le 1$

t의 범위: $x \le t \le 1$

$0 \le x \le t \le 1$

→ 범위 변경 $x: 0 \le x \le t$ $t: 0 \le t \le 1$



$\Rightarrow \int_0^1\int_0^t \cos(t^2)\,dx\,dt = \int_0^1 t\cos(t^2)dt = [\frac{1}{2}\sin(t^2)]_0^1$

$= \frac{1}{2}\sin 1$

$\cos(t^2)$을 상수로 취급

$\therefore \bar{f} = \frac{1}{2}\sin 1$

6

헤비사이드 가리개

$\frac{2}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$ 에서 A,B 구하는 방법

1) A 구하기 — x 가리기, $x=0$ 대입 ⇒ ∴ 2 (A=2)

2) B 구하기 — $(x+1)$ 가리기, $x+1=0$ 대입 ⇒ ∴ −2 (B=−2) $(x=-1)$

$\frac{2}{x} + \frac{-2}{x+1}$

가. $\int(\frac{1}{x+1} - \frac{1}{x-1})dx$

나. $\int(\frac{2}{x} - \frac{2}{x+1})dx$

다. $\int(-\frac{2}{x} + \frac{1}{x^2} + \frac{2}{x+1} + \frac{2}{(x+1)^2})dx$

라. $\int(-\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x+1})dx$

⇒ 나, 다

$\frac{x^2+1}{x^2(x+1)^2}$ 의 경우에는?

$= \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2}$ 에서 큰 차수만 가리기

1) B, D 구하기

$x=0$, $\frac{x^2+1}{(x+1)^2}$ ⇒ $x=0$ 대입 ⇒ B=1, $x=-1$, $\frac{x^2+1}{x^2}$ ⇒ $x=-1$ 대입 ⇒ D=2

∴ $\frac{A}{x} + \frac{1}{x^2} + \frac{C}{x+1} + \frac{2}{(x+1)^2}$ ⇒ 계수 비교 A, C 구하기

→ $Ax(x+1)^2 + (x+1)^2 + Cx^2(x+1) + 2x^2$

x^3은 존재 x ⇒ $A+C=0$ $A=-C$

$x^2+(x+2)x^2=1$ ∴ $C=-2$ $A=2$

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(a) $\sum_{n=1}^{\infty}\frac{n^{\frac{1}{n}}}{n} = \frac{1}{n^{1-\frac{1}{n}}}$ $\lim_{n\to\infty}\frac{\frac{1}{n^{1-\frac{1}{n}}}}{\frac{1}{n}} = 1$ 발산 (∵ L.C.T)

→ 발산 (∵ p-)

(b) $\frac{1}{n^{\sqrt{2}}\ln n} < \frac{1}{2\cdot n^{\sqrt{2}}}$ $\sqrt{2} = 1.414$

$\ln n \triangleq 2$

수렴 (∵ C.T) 수렴 (∵ p-)

(c) $\lim_{n\to\infty}\frac{\tan\frac{1}{n}}{\frac{1}{n}} = 1$ 발산 (∵ L.C.T).

→ 발산 (∵ p-)

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(가)에 의해

$f(x) \overset{\text{Mac}}{=} x + x^2 + \sum_{n=3}^{\infty}\frac{f^{(n)}(0)}{n!}x^n \le x + x^2 + \sum_{n=3}^{\infty}\frac{\frac{n!}{n^2-3n+2}}{n!}x^n$

$\frac{1}{n^2-3n+2} = \frac{1}{n-2} - \frac{1}{n-1}$

$= x + x^2 + \sum_{n=3}^{\infty}(\frac{1}{n-2} - \frac{1}{n-1})x^n$.

$\lim_{n\to\infty}\sum_{k=3}^{n}(\frac{1}{k-2} - \frac{1}{k-1}) = \lim_{n\to\infty}\{(1 - \cancel{\frac{1}{2}}) + (\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}) + \cdots + (\cancel{\frac{1}{n-2}} - \frac{1}{n-1})\}$

$= x + x^2 + x^n$

$\therefore f(1) \le 3 \triangleq m$

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$$\left(\frac{x}{3}+2\right)^n = \sum_{r=0}^{n} \boxed{\left(\frac{x}{3}\right)^r \cdot 2^{n-r}} \cdot {}_nC_r$$

$$\boxed{\left(\frac{x}{3}\right)^r \cdot 2^{n-r}} = 3^{-r}\cdot 2^{n-r}\cdot x^r$$

$r \triangleq n-1 \Rightarrow a_n = {}_nC_{n-1}\cdot 3^{-n+1}\cdot 2^1 = 6n\left(\frac{1}{3}\right)^n$

$\therefore 6\sum_{n=1}^{\infty} n\left(\frac{1}{3}\right)^n. \quad \frac{1}{3}\triangleq x.$

$\sum_{n=1}^{\infty} nx^n = x+2x^2+3x^3+4x^4+\cdots$

$\frac{1}{1-x} = 1+x+x^2+x^3+\cdots$

미분

$\frac{1}{(1-x)^2} = 1+2x+3x^2+4x^3+\cdots$

$\times x$

$\frac{x}{(1-x)^2} = x+2x^2+3x^3+\cdots \Big|_{x=\frac13} = \frac{9}{4}\cdot\frac{1}{3} = \frac{3}{4}$

$\therefore\ 6\cdot\frac{3}{4} = \frac{9}{2}$

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(sol I)

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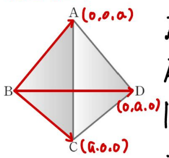

$\vec{AC} = (a,0,-a)$. $\vec{AD} = (0,a,-a)$

$\vec{AB}\times(\vec{AC}\times\vec{AD}) = \langle -a^3, a^3, 0\rangle \Rightarrow \left(-\frac{1}{2\sqrt2}, \frac{1}{2\sqrt2}, 0\right).$

$= \left(-\frac{a}{2}, \frac{a}{2}, 0\right).$

$|\vec{AC}| = 1 \Rightarrow a = \frac{1}{\sqrt2}$

$\therefore$ (ㄷ) $\frac{1}{2}\vec{DC}$ 답

(sol II)

$\vec{AB}\times(\vec{AC}\times\vec{AD}) = (\vec{AB}\cdot\vec{AD})\vec{AC} - (\vec{AB}\cdot\vec{AC})\vec{AD}$

$\vec{AB}\cdot\vec{AD} = |\vec{AB}||\vec{AD}|\cdot\cos\frac{\pi}{3} = \frac12$, $\quad \vec{AB}\cdot\vec{AC} = |\vec{AB}||\vec{AC}|\cdot\cos\frac{\pi}{3} = \frac12$

$= \frac{1}{2}(\vec{AC}-\vec{AD}) = \frac{1}{2}\vec{DC}$

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$x^3+y^3 = 6xy.$

극좌표 $r^3\cos^3\theta + r^3\sin^3\theta = 6r^2\cos\theta\sin\theta.$

$\Rightarrow r = \frac{6\tan\theta\sec\theta}{1+\tan^3\theta}.$

이때, $\tan\theta \triangleq t. \quad \Rightarrow r = \frac{6t\sqrt{1+t^2}}{1+t^3}$

$A = \frac{1}{2}\int_0^{\frac{\pi}{2}} r^2\,d\theta \xrightarrow{\tan\theta\triangleq t} \frac{1}{2}\int_0^{\infty}\frac{36t^2(1+t^2)}{(1+t^3)^2}\cdot\frac{1}{1+t^2}\,dt$

$= 18\int_0^{\infty}\frac{t^2}{(1+t^3)^2}\,dt \xrightarrow{t^3\triangleq u} 6\int_0^{\infty}\frac{1}{(1+u)^2}\,du = 6$

* 데카르트 엽선 *.

· $x^3+y^3 = 3axy \Rightarrow$ 면적 : $\frac{3}{2}a^2$

· $\begin{cases} x = \frac{3at}{1+t^3} \\ y = \frac{3at^2}{1+t^3} \end{cases}$ $(\tan\theta \triangleq t).$

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$\nabla f = \left\langle -\frac{x}{2}, -\frac{y}{2}, 1\right\rangle\Big|_{(a,b,c)} = \left\langle -\frac{a}{2}, -\frac{b}{2}, 1\right\rangle // \langle 1,1,-1\rangle$

$\therefore a=2,\ b=2,\ c=2 \Rightarrow a+b+c=6$

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접평면 : $z = \underbrace{z_x(3,5)}_{=\frac{3}{2}}(x-3)+\underbrace{z_y(3,5)}_{=-\frac{1}{4}}(y-5)+2.$

$x=3.03,\ y=4.98 \Rightarrow z\approx 2.05$

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$g: x^4-x^2+y^4-y^2=0$ & $f=y.$

$\begin{vmatrix} f \\ g \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 4x^3-2x & 4y^3-2y \end{vmatrix} = (2x-4x^3) = 0 \quad \therefore x=0$ or $x=\pm\frac{1}{\sqrt2}$

i) $x=0$: $y^2(y^2-1)=0 \Rightarrow y=-1,0,1$

ii) $x^2=\frac12$: $y^4-y^2-\frac14=0 \Rightarrow y^2 = \frac{1+\sqrt2}{2}$ $(\because y^2>0)$

$\therefore y = \pm\sqrt{\frac{1+\sqrt2}{2}}$

$\therefore M = \sqrt{\frac{1+\sqrt2}{2}}$. $m = -\sqrt{\frac{1+\sqrt2}{2}} \Rightarrow M\cdot m = -\frac{1+\sqrt2}{2}$

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(sol I)

$f_y = \frac{x}{1+(xy)^2} + 4048xye^{y^2-1}$

$\int_0^1\int_0^1 \underbrace{\frac{\partial^2 f}{\partial x\partial y}}_{=f_{yx}}\,dx\,dy = \int_0^1 \underbrace{\left[\frac{x}{1+(xy)^2}+4048xye^{y^2-1}\right]_{x=0}^{x=1}}_{=\frac{1}{1+y^2}-4048ye^{y^2-1}} dy$

$\Rightarrow \left[\tan^{-1}y + 2024e^{y^2-1}\right]_0^1 = \frac{\pi}{4}+2024-\frac{2024}{e}$

(sol II)

$\int_0^1\int_0^1 \underbrace{f_{yx}(x,y)\,dx}_{=\left[f_y(x,y)\right]_0^1 = f_y(1,y)-f_y(0,y)}dy.$

$= \left[f(1,y)\right]_0^1 - \left[f(0,y)\right]_0^1 = f(1,1)-f(1,0)+f(0,1)-f(0,1)+f(0,0)$

16

$x+y \triangleq u,\ y \triangleq v \Rightarrow J(x,y)=1 \Rightarrow J(u,v)=1.$

$\Rightarrow \iint e^{u^2}\,du$

$\underset{\sim\sim}{\text{변환}} \rightarrow (0,0),\ (1,0),\ (1,1)$

$\Rightarrow \int_0^1 \int_0^u e^{u^2}\,dv\,du = \int_0^1 u e^{u^2}\,du = \left[\frac{1}{2}e^{u^2}\right]_0^1 = \frac{1}{2}(e-1)$ //

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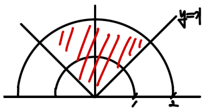

$\rho(x,y) = k\cdot\sqrt{x^2+y^2}$

$\bar{y} = \frac{M_x}{m}$

$m = \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi}\int_1^2 r^2\,dr\,d\theta = \frac{\pi}{2}\cdot\frac{1}{3}\cdot 7 = \frac{7}{6}\pi.$

$M_x = \int_{\frac{\pi}{4}}^{\frac{3}{4}\pi}\int_1^2 r^3\sin\theta\,dr\,d\theta = \underbrace{\left[-\cos\theta\right]_{\frac{\pi}{4}}^{\frac{3}{4}\pi}}_{=\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}}\cdot\frac{1}{4}(16-1) = \frac{15}{4}\sqrt{2}$

$\Rightarrow \bar{y} = \frac{45\sqrt{2}}{14\pi}$ //

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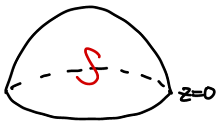

: 폐곡면.

$\iint F\cdot\vec{n}\,dS \overset{\text{발산}}{=} \iiint \underbrace{\operatorname{div}F}_{=2z}\,dV = 2\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^2 \rho\cos\phi\cdot\rho^2\sin\phi\,d\rho\,d\phi\,d\theta.$

$= 4\pi\cdot\left[\frac{1}{2}\sin^2\phi\right]_0^{\frac{\pi}{2}}\cdot\left[\frac{1}{4}\rho^4\right]_0^2 = 2\pi\cdot 4 = 8\pi$ //

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$S \cup S_1 \cup S_2$의 겉넓이!!

$S_2 :\ x^2+y^2 \le 1\ \&\ z=0\ :\ \frac{\pi}{2}.$

$S_1 = \iint\sqrt{1+3}\,dA = 2\boxed{\iint dx\,dy}_{=\frac{\pi}{2}} = \pi$

$S :\ r(\theta,z) = (\cos\theta,\ \sin\theta,\ z)\ \ (0\le\theta\le\pi,\ 0\le z\le\sqrt{3}\sin\theta).$

$= \int_0^{\pi}\int_0^{\sqrt{3}\sin\theta}\underbrace{|r_\theta\times r_z|}_{=1}\,dz\,d\theta = 2\sqrt{3}$

$\therefore$ 겉넓이 $= \frac{3}{2}\pi + 2\sqrt{3}$ //

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$r = \langle 2\sin\phi\cos\theta,\ 2\sin\phi\sin\theta,\ 2\cos\phi\rangle\ \left(0\le\phi\le\frac{\pi}{2},\ 0\le\theta\le 2\pi\right)$

$\int_0^{2\pi}\int_0^{\frac{\pi}{2}} 4\cos^2\phi\cdot\underbrace{|r_\phi\times r_\theta|}_{=4\sin\phi}\,d\phi\,d\theta = 16\cdot 2\pi\left[-\frac{1}{3}\cos^3\phi\right]_0^{\frac{\pi}{2}} = \frac{32}{3}\pi$ //